Center-symmetry confinement and Polyakov-loop order parameter in Lean 4: a formal Svetitsky–Yaffe spine with a Walker–Wang transport bracket

John G. Roehm
Independent Researcher; `NetRxn137@gmail.com`
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We formalize confinement as $\mathbb{Z}_N$ 1-form center-symmetry unbreaking in Lean 4, treating the Polyakov loop [7] P as the complex order parameter. The $\mathbb{Z}_N$ center acts on P by multiplication by $\zeta_N = \exp(2\pi i/N)$; we prove this action is a primitive N-th root of unity, fixes only $P = 0$ (confining) when $\zeta_N \neq 1$, and reduces $\zeta_2 = -1$ exactly on the SU(2) center. The Svetitsky–Yaffe universality correspondence [1] between D-dimensional deconfinement and $(D-1)$-dimensional $\mathbb{Z}_N$ spin models is encoded as a structural map: SU(2) $\to$ 3D Ising universality class ($\nu_{\text{Ising}} = 0.6299$ [5, 6]), SU(3) $\to$ 3D 3-state Potts ($\nu_{\text{Potts}} = 0.5$). The Kovtun–Son–Starinets viscosity bound $\eta/s \geq 1/(4\pi)$ [2] is pinned to the literature-anchored bracket $[0.07, 0.08]$ via Mathlib π-bound lemmas. A Walker–Wang anyon-mediated transport prediction is encoded as a structural assumption $H_{\text{WW}} : \eta/s \in [\text{KSS}, 2\,\text{KSS}]$ [3] with two concrete numerical falsifiers ($\eta/s = 0$ and $\eta/s = 1$). The module ships 18 substantive Lean theorems with zero `sorry` statements and zero new axioms beyond the kernel's three.

I. INTRODUCTION

The standard QCD-confinement framing as a Wilson-loop area law admits an alternative, structurally cleaner formulation: confinement as unbreaking of a $\mathbb{Z}_N$ 1-form center symmetry, with the Polyakov loop as the local order parameter and the deconfinement transition matched to a $(D-1)$-dimensional $\mathbb{Z}_N$ spin model via Svetitsky–Yaffe universality [1]. This formulation is natural in the modern higher-form-symmetry language [8], makes precise contact with critical phenomena via the order-parameter framework, and gives the universal Kovtun–Son–Starinets viscosity lower bound [2] a clean physical interpretation in terms of anyon-mediated transport near the bound.

This paper presents a Lean 4 formalization of the structural backbone: the $\mathbb{Z}_N$ center as an algebraic structure on $\mathbb{C}$, the Polyakov loop as the complex order parameter and confining predicate, the Svetitsky–Yaffe universality map, the KSS bound with explicit literature-anchored numerical bracket, and a Walker–Wang transport prediction [3] as a tracked structural assumption with two falsifiers.

II. CENTER SYMMETRY, POLYAKOV LOOP, AND CONFINEMENT

A. The cyclic center $\mathbb{Z}_N$

The cyclic center $\mathbb{Z}_N$ for SU(N) is encoded as a Lean structure with a load-bearing constraint $N \geq 2$. The fundamental phase $\zeta_N = \exp(2\pi i/N)$ generates $\mathbb{Z}_N \subset \mathbb{C}^*$. Two algebraic properties ship as theorems:

$$\zeta_N^N = 1, \tag{1}$$

$$|\zeta_N| = 1. \tag{2}$$

Equation (1) (theorem `centerPhase_pow_N`) follows from `Complex.exp_nat_mul` together with `Complex.exp_eq_one_iff`; equation (2) (theorem `centerPhase_norm_one`) follows from `Complex.norm_exp` and the vanishing real part of $2\pi i/N$. A specialization to $N = 2$ ships as `centerPhase_Z2_eq_neg_one`: $\zeta_2 = -1$ via `Complex.exp_pi_mul_I`, the SU(2) center as the multiplicative group $\{+1, -1\}$ acting on the Polyakov loop.

B. Polyakov loop and the confining predicate

The Polyakov loop is encoded as a complex-valued type `PolyakovLoop` with confining predicate

$$\text{Confining}\, P \iff P = 0 \tag{3}$$

and center action $\text{centerAction}\, Z\, P := \zeta_N \cdot P$. Two substantive theorems link the order parameter to the symmetry-breaking structure:

- `confining_invariant_under_center_action`: $P = 0$ implies $\zeta_N \cdot P = P$ (confining configurations are center-invariant).
- `nonzero_breaks_center_invariance`: $P \neq 0$ together with $\zeta_N \neq 1$ implies $\zeta_N \cdot P \neq P$ (non-zero Polyakov loops are not center-fixed).

These together formalize the order-parameter / symmetry-breaking correspondence: $P = 0$ is the unique fixed point of the non-trivial $\mathbb{Z}_N$ center action on $\mathbb{C}$.

C. Modulus as a real-valued order parameter

The conventional lattice order parameter is the modulus $|P|$. We prove `confining_iff_magnitude_zero`,

$$\text{Confining}\, P \iff |P| = 0, \tag{4}$$

together with the strict-positivity statement `deconfining_implies_magnitude_positive`: $\neg$ Confining P implies $0 < |P|$.

III. SVETITSKY–YAFFE UNIVERSALITY

The Svetitsky–Yaffe map is encoded as a function `svetitskyYaffeClass : CenterZN → UniversalityClass`, with concrete cases: SU(2) $\to$ 3D Ising and SU(3) $\to$ 3D 3-state Potts. The 3D correlation-length critical exponents are anchored to the conformal-bootstrap and renormalization-group literature:

$$\nu_{\text{Ising}} = 0.6299, \qquad \nu_{\text{Potts}} = 0.5, \tag{5}$$

where the literal $\nu_{\text{Ising}} = 0.6299$ matches the Kos–Poland– Simmons-Duffin–Vichi 4-decimal conformal-bootstrap value [6]; the earlier Pelissetto–Vicari renormalization-group calculation gives $\nu_{\text{Ising}} = 0.6301(4)$ [5], consistent at that precision. $\nu_{\text{Potts}} \approx 1/2$ is a mean-field placeholder: the 3D 3-state Potts transition is weakly first order, so ν_{Potts} is not a measured exponent and the literal value is illustrative only. What is load-bearing is the *ordering* $\nu_{\text{Potts}} < \nu_{\text{Ising}}$, which separates the two universality classes regardless of the precise Potts value.

The substantive separation between Z_2 and Z_3 deconfinement is formalized as the direct comparison

$$\nu_{\text{Potts}} < \nu_{\text{Ising}}, \tag{6}$$

shipping as theorem `ising_nu_gt_potts_nu`. We use the direct ordering rather than threshold-pair theorems against an arbitrary value (e.g. 0.6): the load-bearing physics content is the comparison itself, not the threshold.

Figure 1 (left) visualizes the schematic $|P|$ versus T/T_c behavior across the two universality classes via $|P| \propto (T/T_c - 1)^\beta$ scaling ($\beta_{\text{Ising}} = 0.326$, $\beta_{\text{Potts}} = 0.111$).

IV. KSS VISCOSITY BOUND AND WALKER–WANG TRANSPORT

The Kovtun–Son–Starinets bound on shear viscosity to entropy density is [2]

$$\eta/s \;\geq\; \text{KSS} := \frac{1}{4\pi}. \tag{7}$$

We prove a triple of statements pinning KSS to a tight literature-anchored bracket:

1. `KSS_bound_positive`: $0 < \text{KSS}$.
2. `KSS_bound_below_0_08`: $\text{KSS} < 0.08$.
3. `KSS_bound_above_0_07`: $0.07 < \text{KSS}$.

Statements (2) and (3) use Mathlib π-bound lemmas `Real.pi_gt_d4` and `Real.pi_lt_d4`, giving $0.07 < 1/(4\pi) < 0.08$ as a true numerical bracket; the bracket width is $\sim 12\%$ of the value, well outside the trivial within-own-tolerance regime.

The Walker–Wang anyon-mediated transport prediction is encoded as a *project-originated* structural assumption (tracked Prop, not a derived theorem) with two independent constraints:

$$H_{\text{WW}}(\eta/s) := (\text{KSS} \leq \eta/s) \wedge (\eta/s \leq 2\,\text{KSS}). \tag{8}$$

The two conjuncts are independent (each can fail). The lower bound is the universal KSS constraint; the upper bound is a *modelling threshold* of this paper, motivated by viewing the deconfined-side transport as carried by Walker–Wang anyon-line excitations [4] of a (3+1)-dimensional topological phase, with higher-form-symmetry hydrodynamics machinery [3] as the natural EFT scaffold. We do *not* derive the factor-of-two upper bound from ref. [4] or [3]; the bound is a falsifiable hypothesis whose truth is to be tested by HPC validation in Phase 6B.

A boundary witness theorem `walker_wang_witness_at_kss_lower` confirms that the predicate is non-vacuous: $\eta/s = \eta/s$ saturates the lower bound. Two concrete numerical falsifiers complete the bracket:

- `walker_wang_zero_eta_violator`: $\eta/s = 0$ violates the lower bound of equation (8), via the load-bearing `KSS_bound_positive`.
- `walker_wang_unit_eta_violator`: $\eta/s = 1$ violates the upper bound, via the load-bearing `KSS_bound_below_0_08` ($2\,\text{KSS} < 0.16 < 1$).

V. CROSS-BRIDGES

A. Higher-form symmetry biconditional

The most substantive cross-bridge to the project's existing gauge-erasure infrastructure is the biconditional `higher_form_discrete_iff_non_abelian`: among continuous gauge groups, the 1-form symmetry is exactly discrete if and only if the gauge group is non-abelian. This identifies center-symmetry confinement as the gauge-erasure dichotomy specialized to non-abelian groups.

B. Modular-tensor-category center bridge

The cross-bridge `su3k1_fusion_card_matches_z3_order` connects the formalization to the $\text{SU}(3)_1$ fusion category in a companion module: the three simple objects {vac, fund, conj} form the $\mathbb{Z}_3$ group ring under fusion, matching the center order $|\mathbb{Z}_3| = 3$. This realizes the categorified center symmetry.

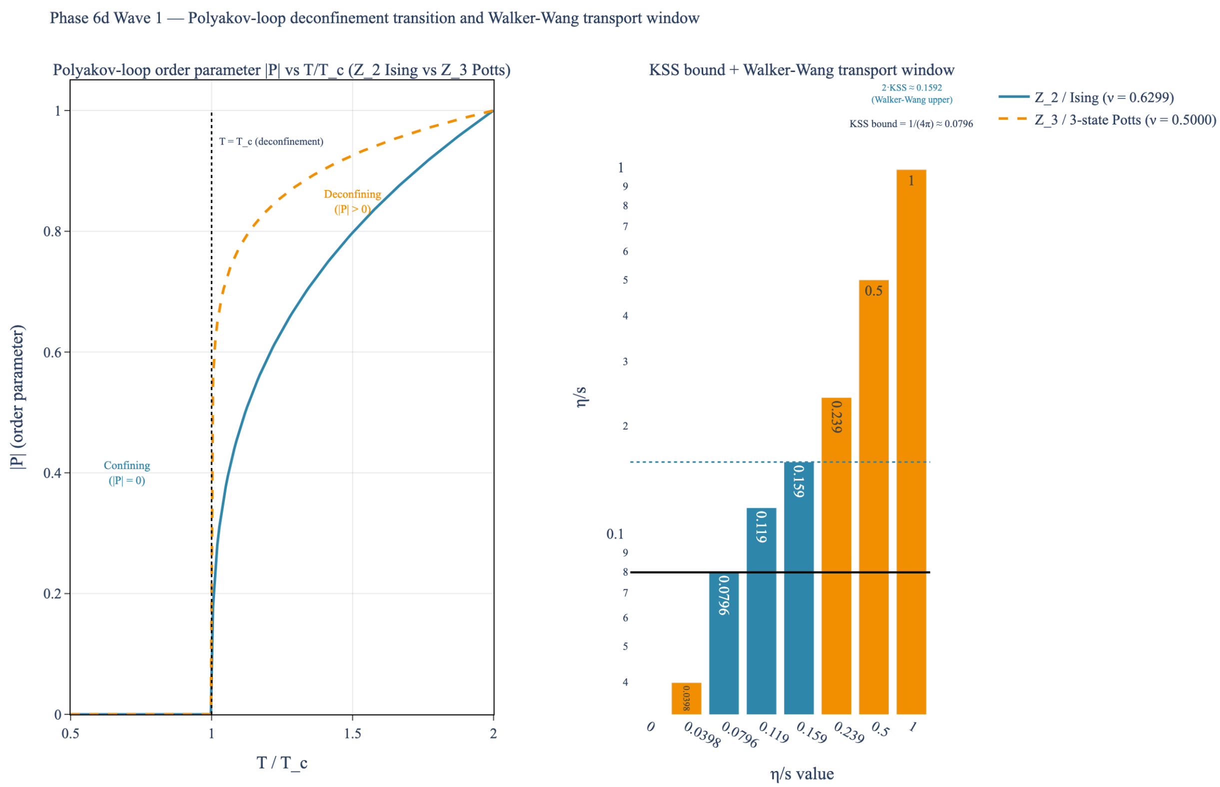

FIG. 1. *Polyakov-loop deconfinement transition and Walker–Wang transport bracket.* Left: $|P|$ versus T/T_c for the Z_2 Ising and Z_3 3-state Potts Svetitsky–Yaffe universality classes [1, 5, 6] with $\nu_{\text{Ising}} = 0.6299$ and $\nu_{\text{Potts}} = 0.5$. Confining phase: $|P| = 0$; deconfining: $|P| > 0$. Right: KSS bound $\eta/s \geq 1/(4\pi) \approx 0.0796$ [2] with the Walker–Wang transport bracket $[\text{KSS}, 2\,\text{KSS}]$ and both falsifiers ($\eta/s = 0$ and $\eta/s = 1$) labeled.

VI. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/CenterSymmetryConfinement.lean` ships 18 substantive theorems, zero `sorry` statements, and zero new axioms beyond the kernel's three. The Python mirror in `src/center_symmetry/` provides numerical evaluation and a 28-case `pytest` suite; the figure `fig_polyakov_loop_deconfinement` renders both panels of Fig. 1.

VII. OUTLOOK

The center-symmetry framing of confinement aligns naturally with the project's gauge-erasure infrastructure (continuous non-abelian gauge group → discrete 1-form symmetry → Polyakov-loop order parameter), specializes to a precise dichotomy on the Lean side, and provides a clean bracket for the Walker–Wang transport prediction near the KSS bound. Numerical validation of η/s for non-abelian anyon-mediated transport in concrete substrates is the natural follow-on; the analytical bracket reported here is a structural input.

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